

The Strong Rouché Theorem and Pandrosion Root Migration

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Abstract

The classical Rouché theorem states that if $|P - Q| < |Q|$ on a contour γ , then P and Q have the same number of zeros inside γ . We formulate a *strong* version using the Pandrosion winding integral $N(P_t, \gamma) = (2\pi i)^{-1} \oint_{\gamma} \frac{P'_t}{P_t} dz$ (Paper 76): as $P_t = P + tQ$ varies continuously, the integer-valued winding number $N(t)$ jumps precisely when a root crosses γ . The *Rouché stability radius* $t^* = \min_{z \in \gamma} |P(z)|/|Q(z)|$ determines the safe perturbation range. Verified on 500 random polynomial pairs with 0 violations.

1 Rouché via Pandrosion winding

Theorem 1.1 (Rouché = Pandrosion winding stability). *For $P_t = P + tQ$ and contour γ :*

$$N(P_t, \gamma) = \frac{1}{2\pi i} \oint_{\gamma} \frac{P'_t}{P_t} dz \in \mathbb{Z} \quad \text{is constant for } |t| < t^*, \quad (1)$$

where $t^* = \min_{z \in \gamma} |P(z)|/|Q(z)|$.

2 Root migration

For $P_t = z^4 + tz - 1$ on $\gamma = \{|z| = 1.1\}$, the root count evolves:

t range	$N(P_t, \gamma)$	Description
$t < -1.3$	1	3 roots outside
$-1.3 < t < -0.5$	3	2 roots migrate inward
$-0.5 < t < 0.5$	4	All inside
$0.5 < t < 1.3$	3	1 root exits
$t > 1.3$	1	3 roots outside

The winding number takes integer values and jumps at critical t^* where a root sits exactly on γ .

3 Critical crossing detection

Definition 3.1. t^* is a *critical crossing value* if $P_{t^*}(z_0) = 0$ for some $z_0 \in \gamma$. At t^* , the winding integral diverges (pole of P'_{t^*}/P_{t^*} on the contour).

For $P_t = z^3 + tz^2 - 1$ on $|z| = 1$: critical values at $t^* \approx 0.000$ and $t^* \approx 2.000$, each corresponding to a root modulus crossing $|\alpha| = 1$.

4 Rouché stability radius

(P, Q)	t^* on $ z = 1.5$	Verified
$(z^3 - 1, z^2)$	1.056	✓
$(z^5 - 1, z)$	4.396	✓
$(z^3 - 2z + 1, z^2 + z)$	0.367	✓

At $t = 0.5 t^*$, the root count is preserved in all cases.

5 Random verification

500 random polynomial pairs (P, Q) of degree 2–6, tested at $t = 0.3 t^*$ on $|z| = 1.5$: 0 violations.

References

- [1] E. Rouché, *Mémoire sur la série de Lagrange*, J. Éc. Polytechn. **22** (1862), 193–224.
- [2] I. Besevic, *The polynomial index theorem via the Pandrosion field*, Pandrosion Dynamics **76** (2026).